

Assignment 5 - Machine Learning

Theoretical Questions

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Question 1

Limitations:

- Assumes spherical and convex shapes for each cluster.
- Requires specification of number of clusters.
- Gives poor results when initialized badly.

Initialization Methods:

- **Random:** Randomly samples K distinct data points as centroids.
- **K-Means++:** Selects the first centroid randomly, and then each subsequent centroid is chosen from the remaining points with a probability proportional to its squared distance from the nearest existing centroid.

Can K-Means be used with categorical data? Not directly, but it's possible if we encode the categorical features using a feature encoding method such as *Label Encoding*.

Question 2

It could be due to setting an small ϵ .

Question 3

- a) Because it doesn't assign a given data point to a certain cluster, instead it assign the data point to each of the clusters, with a probability, but hard clustering methods do the opposite.
- b) Yes. Below is the explanation for anomaly detection using each of the methods:
- **DBSCAN:** At the end of algorithm's execution, the points that have not been assigned to any clusters, are anomalies.
 - **GMM:** The data points with a very low likelihood of occurrence, after running some iterations of the algorithm, are considered as anomalies.

Question 4

- a) Their difference is written below. Also we can't use **Silhouette Score** for a dataset which includes only one cluster, since one of parameters in the computation is the average distance from a given point and all other points of the nearest neighbor cluster.
- **Silhouette Score:** Computationally inefficient - point-wise calculation of score - higher value is better.
 - **Davis-Bouldin Score:** Computationally efficient - cluster-wise calculation of score - lower value is better.
- b) When having well-separated clusters, the silhouette score will be close to 1, and as the clusters tend to overlap with each other, the score decreases to -1.

Question 5

- a)
- t-SNE is more computationally expensive than PCA.
 - t-SNE can capture non-linear patterns but PCA can only capture linear patterns.
 - t-SNE is primarily used for visualization in 2D or 3D space but PCA could also be used for feature reduction.

- b) Yes that's possible. e.g. we can first apply PCA for feature reduction and then t-SNE for visualization.
- c)
- Dynamic Time Warping (DTW) is the most appropriate distance metric for clustering time series data because it can compare sequences that vary in speed or alignment.
 - DTW works by non-linearly warping the time axis of one sequence to optimally align it with another, minimizing the cumulative distance between their matched points.
 - Unlike Euclidean distance, which assumes fixed one-to-one correspondence between time steps, DTW accounts for time shifts and distortions, making it more robust for real-world temporal data.
 - As a result, DTW is well-suited for clustering time series where similar patterns may be temporally misaligned but structurally similar.

Question 6

$$X = \begin{bmatrix} 90 & 60 & 90 \\ 90 & 90 & 30 \\ 60 & 60 & 60 \end{bmatrix}$$

Step 1: Center the Data

Compute the mean of each column:

$$\mu = \frac{1}{3} \sum X = \begin{bmatrix} \frac{90+90+60}{3} \\ \frac{60+90+60}{3} \\ \frac{90+30+60}{3} \end{bmatrix} = \begin{bmatrix} 80 \\ 70 \\ 60 \end{bmatrix}$$

Subtract the mean vector from each row:

$$X_{\text{centered}} = X - \mu^T = \begin{bmatrix} 90 - 80 & 60 - 70 & 90 - 60 \\ 90 - 80 & 90 - 70 & 30 - 60 \\ 60 - 80 & 60 - 70 & 60 - 60 \end{bmatrix} = \begin{bmatrix} 10 & -10 & 30 \\ 10 & 20 & -30 \\ -20 & -10 & 0 \end{bmatrix}$$

Step 2: Compute the Covariance Matrix

$$\Sigma = \frac{1}{n-1} X_{\text{centered}}^T X_{\text{centered}}$$

$$\Sigma = \frac{1}{2} \begin{bmatrix} 10 & 10 & -20 \\ -10 & 20 & -10 \\ 30 & -30 & 0 \end{bmatrix}^T \begin{bmatrix} 10 & 10 & -20 \\ -10 & 20 & -10 \\ 30 & -30 & 0 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1000 & -500 & 600 \\ -500 & 600 & -900 \\ 600 & -900 & 1800 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 500 & -250 & 300 \\ -250 & 300 & -450 \\ 300 & -450 & 900 \end{bmatrix}$$

Step 3: Find Eigenvalues and Eigenvectors

Solve the characteristic equation:

$$\det(\Sigma - \lambda I) = 0$$

This yields eigenvalues:

$$\lambda_1 = 1205.47, \quad \lambda_2 = 336.20, \quad \lambda_3 = 158.32$$

Corresponding eigenvectors (normalized):

$$v_1 = \begin{bmatrix} 0.35 \\ -0.49 \\ 0.80 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0.84 \\ 0.54 \\ -0.05 \end{bmatrix}$$

Step 4: Project the Data onto the First Two Principal Components

Let $W = [v_1 \ v_2]$. Then the reduced data:

$$X_{\text{reduced}} = X_{\text{centered}} \cdot W$$

$$X_{\text{reduced}} = \begin{bmatrix} 10 & -10 & 30 \\ 10 & 20 & -30 \\ -20 & -10 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0.35 & 0.84 \\ -0.49 & 0.54 \\ 0.80 & -0.05 \end{bmatrix} = \begin{bmatrix} 30.5 & 0.6 \\ -30.7 & 14.6 \\ 0.2 & -15.2 \end{bmatrix}$$

Conclusion

The data projected into the 2D space defined by the first two principal components is:

$$\begin{bmatrix} 30.5 & 0.6 \\ -30.7 & 14.6 \\ 0.2 & -15.2 \end{bmatrix}$$

This concludes the PCA procedure.

Question 7

Clustering may benefit **Linear Regression** in two ways:

- Detection of anomalies and removing them before applying **Linear Regression**.
- Applying clustering in order to divide the data into multiple group and applying **Linear Regression** on each separately.

Question 8

This shows that each data point is at an equal distance to all of its other neighbors and all are within a single cluster.

Consider this scenario: We have three clusters A, B, and C (each with a single data point in them). The distances AB, AC, and BC are 4, $\sqrt{13}$, and $\sqrt{13}$ respectively. Now if after merging A and B together, we get a centroid for that cluster that is of distance 3 from cluster C, the distance has been reduced and this is distance inversion.